Combustion and Flame 204 (2019) 162–175

Contents lists available at ScienceDirect

Combustion and Flame

journal homepage: www.elsevier.com/locate/combustflame

Measurement and modelling of the laminar burning velocity of methane-ammonia-air flames at high pressures using a reduced reaction mechanism

Ekenechukwu Chijioke Okafor*, Yuji Naito, Sophie Colson, Akinori Ichikawa, Taku Kudo, Akihiro Hayakawa, Hideaki Kobayashi

Institute of Fluid Science, Tohoku University, 2-1-1 Katahira, Aoba-Ku, Sendai 980-8577, Japan

ARTICLE INFO

Article history:
Received 19 October 2018
Revised 11 December 2018
Accepted 7 March 2019

Keywords:
Methane
Ammonia
Laminar burning velocity
Markstein length
Reduced mechanism

ABSTRACT

Ammonia and blends of ammonia with methane are gaining increased interest as fuels for gas turbine applications and hence optimized reduced reaction mechanisms for the fuels are required for the development of combustors. However, there is a scarcity of measured data on the laminar burning velocity of the fuels for optimizing and validating reaction mechanisms especially at high pressures. In this study, an extensive set of measurements of the unstretched laminar burning velocity and Markstein lengths of CH_4–NH_3-air flames at high pressures are reported for the first time and an optimized reduced reaction mechanism is proposed. The experiments were conducted in a constant volume chamber for various ammonia heat fractions in the fuel ranging from 0 to 0.30, equivalence ratios ranging from 0.7 to 1.3, and mixture pressures ranging from 0.10 to 0.50 MPa. The reduced reaction mechanism was developed from a detailed reaction mechanism for CH_4–NH_3-air flames by Okafor et al., and optimized against the present measurements and data in the literature. It is shown that the reduced mechanism models the unstretched laminar burning velocity of NH_3/air and CH_4–NH_3-air flames with high fidelity at all studied conditions. It also models satisfactorily NH_3, NO and CO concentration in CH_4–NH_3–O_2–N_2 oxidation in a laminar flow reactor. Furthermore, the reduced mechanism is demonstrated to predict NO, OH, and NH profiles in a premixed stagnation NH_3-air flame satisfactorily. The experimental measurements were also used to validate selected detailed reaction mechanisms. It was found that the significant over-prediction of NO production from NH_3 oxidation by GRI Mech 3.0 is primarily due to the influence of the reaction $NH+H_2O=HNO+H_2$, which in fact may not be important in fuel NO chemistry.

1. Introduction

Ammonia is a carbon-free molecule which has attracted interest as a hydrogen carrier and a fuel for combustion systems, especially micro gas turbines and industrial furnaces [1–5]. Ammonia has a high hydrogen density and, having boiling and condensation temperatures similar to propane, it is easier to store and transport than hydrogen thus making it a promising hydrogen carrier. Ammonia can be produced from renewable sources and, being one of the globally most produced chemicals with a mass production history of more than 100 years, it has a well-established production and distribution network.

Ammonia-air mixtures however have a relatively low burning velocity, which is about five times lower than that of CH_4-air at stoichiometric condition at 298 K and 0.10 MPa. Pratt [6] reported that the low burning velocity inhibits the use of high inlet velocities to achieve effective turbulent mixing in ammonia gas turbine combustors because the fluid residence time in the combustor would be too short for complete combustion of ammonia. The use of low inlet velocities however inhibits turbulent mixing and flame stabilization and thus may lead to a low combustion efficiency. Furthermore, ammonia contains a nitrogen atom in its molecule. Consequently, ammonia combustion may lead to the production of large amounts of nitric oxides through the fuel NOx chemistry. Ammonia-fuelled micro gas turbines may emit NOx in the order of 1000 ppmv [1]. These challenges of ammonia combustion contributed to ammonia being seldom considered a fuel in the past. A vast body of literature on ammonia oxidation chemistry however abound, motivated mostly by the need to understand the chemistry of fuel NOx formation [4,7–10].

A renewed interest in ammonia application as a fuel has however recently been encouraged especially in micro gas turbines by

* Corresponding author.
E-mail address: okafor@flame.ifs.tohoku.ac.jp (E.C. Okafor).

https://doi.org/10.1016/j.combustflame.2019.03.008

recent experimental and numerical studies which propose methods of overcoming the challenges of the low burning velocity and the high fuel NOx production [1–4,11–13]. These studies have shown that the application of strong swirling flows in ammonia gas turbine combustors enhance flame stability while NOx emission from the combustor can be controlled through two-stage rich-lean combustion strategy. Okafor et al. [2] demonstrated that a control of equivalence ratio upstream of a two-stage premixed ammonia micro gas turbine combustor resulted to NOx emissions as low as 42 ppm with a combustion efficiency of 99.5%.

In addition, there is growing interest in the application of blends of ammonia and other fuels such as hydrogen [14–17], diesel [18,19], and methane [1,20,21] in combustion devices. Blends of ammonia and hydrocarbons may result to low-carbon fuels with flame speeds higher than that of ammonia. Hence, such blending offers potentials for an improvement of the combustion characteristics of ammonia mixtures and a reduction in CO_2 emission from hydrocarbon combustion. Kurata et al. [1] studied the application of CH_4–NH_3 binary fuel in a 50 kW micro gas turbine and reported that the use of the binary fuel resulted in higher combustion and thermal efficiencies than the case of pure ammonia fuel. However, the binary fuel produced significantly higher levels of NOx emissions than pure ammonia in the micro gas turbine.

For the development of combustors for these new fuels, accurate prediction of the laminar burning velocity of the flames and NOx formation/reduction are important. The laminar burning velocity is one of the most important global properties of a fuel which could be used to characterize many premixed flame phenomena. The characteristic chemical time, and hence the Damkholer number in a combustor is a function of the laminar burning velocity. Therefore, accurate prediction of flame stabilization, flame height and consequently emissions of unburned NH_3 and NOx in ammonia-fuelled two-stage gas turbine combustors may be dependent on accurate modelling of the laminar burning velocity [2,12]. Similarly, accurate model prediction of the optimum low-NOx upstream equivalence ratio in two-stage combustors fuelled with ammonia mixtures [2], which may be an important combustor design parameter, requires an accurate prediction of the burning velocity as well as NOx emissions from the flame.

Numerous hydrocarbon oxidation mechanisms which have ammonia oxidation sub-mechanism have been well optimized and validated for predicting the burning velocity of hydrocarbon flames such as methane flames with some degree of accuracy. However, their ammonia oxidation kinetics have not been well optimized for modeling flames with high ammonia concentrations, especially NH_3-air flames. The recent review of ammonia oxidation mechanisms by Kobayashi et al. [4] indicates that wide disparities exist in the prediction of the laminar burning velocity of NH_3-air flames by the existing detailed reaction mechanisms. Figure 1 is a comparison of measured burning velocity of NH_3-air flames with prediction by a select detailed reaction mechanisms. The reaction mechanisms of Glarborg [10], Konnov [35], Miller [30], and Lindstedt [32] significantly over-predict the measurements. GRI Mech 3.0 [36] is among the detailed reaction mechanisms that model the measurements in Fig. 1 closely. It also models the burning velocity of CH_4–NH_3-air flames at 0.10 MPa satisfactorily [20]. However, GRI Mech 3.0 [36] was developed and optimized for natural gas combustion and lacks important NH_3 oxidation steps, thus may not model satisfactorily NO formation/reduction in the oxidation of mixtures with high ammonia concentration. On the other hand, Tian's mechanism [28] which also models the measurements in Fig. 1 satisfactorily is a detailed CH_4–NH_3 oxidation kinetics optimized with measured species profiles in stoichiometric flames of NH_3–CH_4–O_2-Ar. The mechanism has also been reported to predict NOx emissions from CH_4–NH_3-air flames satisfactorily [38]. However, it largely underestimates the unstretched laminar burning velocity of CH_4–NH_3-air flames at 0.10 MPa owing mainly to the dominance of HCO (+H, OH, O_2) = CO (+ H_2, H_2O, HO_2) over HCO = CO + H in the conversion of HCO to CO [21]. The mechanism by Okafor et al. [21] is a detailed reaction mechanism based on GRI Mech 3.0 [36] with the addition of some important ammonia oxidation chemistry from Tian's mechanism [28]. The mechanism was optimized with measured unstretched laminar burning velocity of CH_4- air and CH_4–NH_3-air flames at 0.10 MPa and models the burning velocity of NH_3-air flames very closely as shown in Fig. 1.

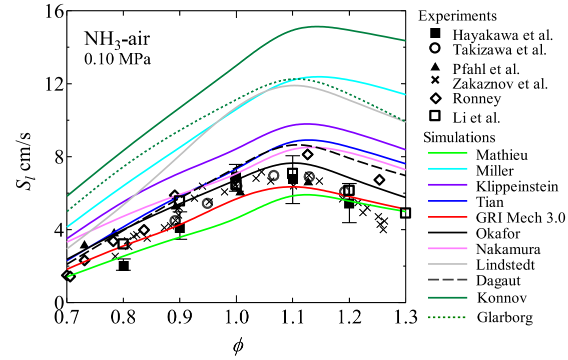

Fig. 1. Comparison of the measured burning velocity of NH_3-air flames by Hayakawa et al. [22], Takizawa et al. [23], Pfahl et al. [24], Zakaznov et al. [25], Ronney [26] and Li et al. [27] to predictions made by the kinetics models of Glarborg [10], Okafor [21], Tian [28], Miller [30], Mathieu [31], Lindstedt [32], Klippenstein [33], Dagaut [34], Konnov [35], GRI Mech 3.0 [36], and Nakamura [37], updated from [4].

Further optimization and validation of the detailed reaction mechanisms, especially at conditions more relevant to gas turbine applications such as elevated pressure conditions are necessary. However, there is a scarcity of experimental measurement of laminar burning velocity of ammonia mixtures at high pressures. Hayakawa et al. [22] recently measured for the first time the unstretched laminar burning velocity of NH_3-air flames at high pressures up to 0.50 MPa. They compared their measurements at stoichiometric condition with model predictions by a series of detailed reaction mechanisms and concluded that the reaction mechanisms do not model the burning velocity satisfactory at high pressures. On the other hand, to the best of our knowledge, there is yet no reported measurements of the laminar burning velocity of CH_4–NH_3-air flames at high pressures up to 0.50 MPa. This is one of the focus of the present study.

On the other hand, reduced reaction mechanisms, with fewer number of species and reactions than the detailed mechanism, that can satisfactorily reproduce the relevant physical phenomena under targeted conditions are desirable in the development of combustors [39]. Large-scale or 3D combustion modelling with detailed reaction mechanisms involves huge computational costs in terms of memory and computation time. The use of detailed reaction mechanisms is further limited by the problem of stiffness caused by large disparities in timescale for evolution of different species [40]. Hence, most 3D LES modelling of NH_3-air flames [11–13] employ the mechanism of Miller et al. [30] (22 species, 92 elementary reactions) because of its smaller size compared to other ammonia oxidation mechanisms. However, the mechanism by Miller et al. [30] is only applicable to pure ammonia flames, and it significantly over estimates the unstretched laminar burning velocity of NH_3-air flames. Most reaction mechanisms for CH_4–NH_3 combustion are significantly large. Consequently, there is a demand for reduced chemical kinetic mechanisms that can reproduce the flame speed and NOx concentrations of NH_3-air and CH_4–NH_3 flames.

Therefore, in the present study, an extensive set of new experimental data on the unstretched laminar burning velocity of

Table 1
Experimental conditions.

ϕ	P_i MPa	E_{NH_3}**
0.7	0.1	0, 0.1, 0.2, 0.3
0.8	0.3, 0.5	0, 0.1, 0.2, 0.3
0.9	0.1, 0.3, 0.5	0, 0.05, 0.1, 0.2, 0.3
1.0	0.1, 0.3, 0.5	0, 0.05, 0.1, 0.2, 0.3
1.1	0.3, 0.5	0, 0.1, 0.2, 0.3
1.2	0.3, 0.5	0, 0.1, 0.2, 0.3
1.3	0.3, 0.5	0, 0.1, 0.2, 0.3

** $E_{NH_3} = (x_{NH3}LHV_{NH3})/(x_{NH3}LHV_{NH3} + x_{CH4}LHV_{CH4})$, where, x_{NH_3} and x_{CH_4} are mole fractions of NH_3 and CH_4 in the binary fuel and LHV stands for the lower heating value (LHV_{CH4}= 802.30 kJ/mol, LHV_{NH3}= 316.84 kJ/mol).

CH_4–NH_3-air flames are reported and a reduced reaction mechanism for modelling NH_3-air and CH_4–NH_3-air flames is proposed, which is optimized and validated with results of present measurement and data from the literature. Furthermore, the burned gas Markstein length of the flames are also reported in this study. The Markstein length or the Markstein number quantifies the effects of flame stretch rate on the flame speed, thus it is relevant in turbulent flame modelling. It is the preferred parameter to the Lewis number in the modelling of turbulent flames [41–43]. The Markstein length is also important in characterizing the propensity of cellular flame instability [44,45]. However, data on the Markstein length of CH_4–NH_3-air flames at elevated pressures have not been reported to date.

2. Details of the experiments

The constant volume chamber and the experimental setup used in this study are described in our previous articles [16,21,22]. The combustion chamber is cylindrical and has sufficiently large inner wall to optical window diameters ratio of 270 mm/60 mm. Mixtures of methane (99.9% purity), ammonia (>99.9% purity) and dry air from a laboratory air compressor with a drier were prepared in the chamber according to the required partial pressures of the gases using a GE UNIK 5000 silicon pressure sensor with an enhanced premium accuracy of 0.02 FS and a full scale range of 0.50 MPa. The mixtures were ignited at a point along the central axis of the cylindrical chamber using capacitor discharge ignition (CDI). The diameter of the spark electrodes was 1.5 mm and the spark gap was set to 1 mm. The electrostatic energy, which was charged in the capacitor in the CDI circuit, was set to 0.3 J. Flame propagation up to a diameter of 60 mm was observed and recorded by Schlieren photography via the quartz windows of the chamber with a high-speed digital camera (Photron, FASTCAM Mini UX100), while the pressure rise inside the chamber during flame propagation was measured using a Kistler pressure transducer. The pixel setting for the camera was 768 $\times$ 768 and the framing rates adopted ranged from 2000 to 4000 f/s, depending on the flame propagation speeds.

The experimental conditions in the present study are shown in Table 1. The mixture temperature, T_u, was 298 $\pm$3 K while the mixture pressure, P_i, varied from 0.10 to 0.50 MPa for different ammonia concentrations in the fuel expressed in terms of the heat fraction, E_{NH_3}. For the equivalence ratio, ϕ , of 0.7, experiments were conducted only at 0.10 MPa, while for the equivalence ratios of 0.9 and 1.0, experiments were conducted at 0.10 MPa only for the mixtures with E_{NH_3}= 0.05. These measurements at 0.10 MPa are complementary to the previously reported data of CH_4–NH_3-air flames at 0.10 MPa [21].

The value of E_{NH_3} was varied from 0 to 0.3, corresponding to a variation in x_{NH_3} from 0 to 0.52 at all equivalence ratios. The equivalence ratio was calculated considering the following oxidation reaction,

$$nCH_4 + mNH_3 + (2n + 0.75m)(O_2 + 3.76N_2) = nCO_2 + (2n + 1.5m)H_2O + 3.76(2n + 0.75m)N_2 \quad (1)$$

Note that N_2 rather than NO is the final product of ammonia combustion when considering the equilibrium products of the reaction. Even though ammonia is associated with high NO production in the flame, the equilibrium NO concentration in NH_3-air combustion is lower than that in CH_4-air combustion [4]. In this study, at least five experimental runs were done at each examined condition.

Several uncertainty factors can be associated with measurement of the unstretched laminar burning velocity using the constant pressure spherical flame technique as discussed by Egolfopoulos [46], Chen [47], and in a recent review by Konnov et al. [48]. These factors were carefully addressed in the present measurements. The most significant sources of uncertainty in the present measurements were associated with radiation heat loss, and preparation of the multicomponent fuel mixtures. The fuel-independent correlation proposed by Yu et al. [49] was used to estimate the uncertainty due to radiation. Radiation induced uncertainty was within 5% and 2% at 0.10 MPa for mixtures with unstretched laminar burning velocities above 12 cm/s and 26 cm/s, respectively. The effect of radiation increases with an increase in mixture pressure due to a decrease in flame propagation speed [50], reaching about 11% for the slowest burning flame at 0.5 MPa. A GE pressure transducer with an enhanced premium accuracy of 0.02% FS and full scale range of 0.5 MPa was used in mixture preparation. Therefore, the uncertainty in mixture preparation was highest at atmospheric pressure, and decreased significantly as the accuracy of the pressure measurement increased with mixture pressure. Other potential sources of uncertainty in the measurement include confinement, flame instability, buoyancy, ignition energy, nonlinearity of flame speed-stretch rate relationship, etc. [47]. However, the effects of these factors are negligible in the present measurements as explained below.

It has been shown that there is no pressure rise during the period of recording of flame propagating in this chamber, owing to the sufficiently large ratio of the inner wall diameter to the optical window diameter [16,21,22]. This was also confirmed in the present measurements at high pressures, indicating that confinement has no effects on the measured data in this study. Hence, the measurements were considered to have been completed at the adopted initial mixture pressure and temperature. Likewise, there is no significant influence of flame instability in the reported data. Figure 2 shows the flame images of the lean CH_4-air flame at 0.50 MPa which showed the highest propensity to be unstable of all the flames studied. Cracks originating from the effect of the spark and the deformation of the flame by the electrodes can be observed on the flame front. Propagation of the cracks may result to the development of a cellular structure which may enhance the flame speed due to an increase in the flame area. In the present study however, the cracks did not propagate into an unstable flame structure within the recorded range of flame radii. The mere presence of cracks on the flame front could lead to uncertainty in the measured flame speed, however the effect is negligible and not easily quantifiable [51–54]. Furthermore, buoyancy may lead to a severe deformation of the shape of slow-burning outwardly-propagating flames, leading to errors in the measured flame speed [22,55]. Figure 3 shows the flame of E_{NH_3} = 0.3 and ϕ = 1.3 at 0.50 MPa which was the most susceptible flame to buoyancy effects because it had the lowest flame propagation speed. An upward displacement of the flame from the point of ignition can be observed, however there is no appreciable deformation of the flame shape within the recorded range of flame radii. Therefore, even though the upward displacement of the flame may have

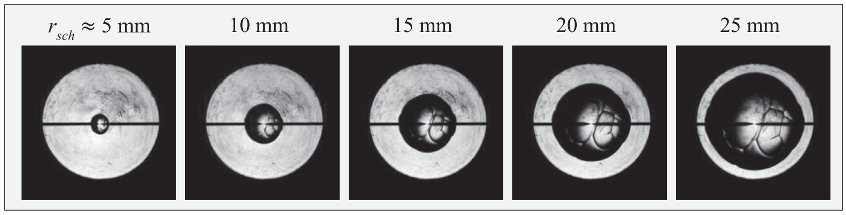

Fig. 2. Propagation sequence of the flame most susceptible to flame instability due to hydrodynamic and thermo-diffusive effects in the present study (ϕ = 0.8, $E_{NH_3} = 0$, $P_i = 0.50$ MPa).

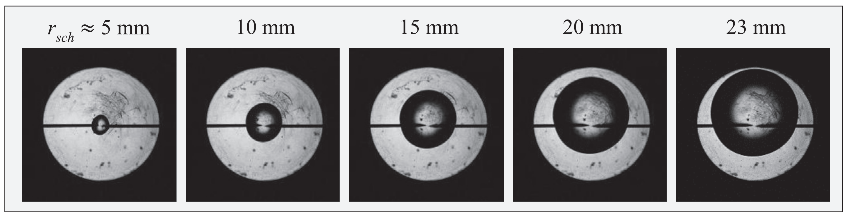

Fig. 3. Propagation sequence of the flame most susceptible to buoyancy effects in the present study ($\phi = 1.3$, $E_{NH_3} = 0.3$, $P_i = 0.50$ MPa).

some effects on the flame speed, the effect is considered insignificant and the overall effect of buoyance is considered not important for the range of flame radii used in evaluating the flame speed in the present study. Likewise, the effect of ignition energy was considered to have been avoided in the reported results because the minimum flame radius used in extracting the unstretched laminar burning velocity and the Markstein length was 10 mm. Furthermore, non-linear stretch extrapolation method, explained in more details in the following paragraphs, was found suitable in this study to minimize uncertainties arising from stretch extrapolation. It has been shown that the disparity between the linear and non-linear extrapolations increase with an increase in equivalence ratio and ammonia concentration in CH_4–NH_3-air flames at atmospheric pressure [21]. This is attributed to an increasing non-linear behavior of the flame speed-stretch rate relationship as the sensitivity of the flame to stretch increases, consequently reducing the accuracy of the linear extrapolation. Unlike the non-linear method used here, the liner method is largely sensitive to the range of flame radii used in the extrapolation [47]. Huo et al. [56] however showed that the use of large initial and final radii in the stretch extrapolation may reduce the disparity in the results of linear and non-linear methods. Another potential source of uncertainty is associated with the high solubility of ammonia in water. During the experiments, a significant amount of water vapor condensed inside the chamber after each experimental run. This was promoted with an increase in ammonia concentration in the fuel and with an increase in mixture pressure. Therefore, the chamber was dried after each run to avoid uncertainties in mixture fraction due to absorption of ammonia molecules by condensed water vapor in the chamber. Furthermore, ammonia is known to adsorb on stainless steel [57]. This may contribute to a significant uncertainty in mixture preparation for mixtures with very low ammonia concentration and long residence time in a stainless steel chamber, and if the surface area-to-volume ratio of the chamber is sufficiently large. Consequently, it may be necessary to passivate the chamber walls with ammonia to reduce ammonia adsorption [31]. However, for the mixture with the lowest ammonia concentration in the present study ammonia adsorption on the chamber walls is estimated to contribute to less than 0.4% reduction in ammonia concentration, which has negligible effect on the mixture fraction and equivalence ratio.

The images of the propagating spherical laminar flames obtained from the experiment were processed to obtain the flame speed by measuring the radius of the flames, r_{sch}, with time, t. The laminar flame speed, s_n, was then evaluated as the derivative of the flame radius with respect to time. Because spherically propagating flames are affected by stretch due to curvature and/or aerodynamic strains, the flame speed is dependent on the total flame stretch rate, ε, of the spherical flame defined by Eq. (2) [58].

$$\varepsilon = \frac{1}{A_f}\frac{dA_f}{dt} = \frac{2}{r_{sch}}\frac{dr_{sch}}{dt} \tag{2}$$

where A_f is the area of the spherical flame front. Representative data of the measured flame radius with time, laminar flame speed and total flame stretch rate for each experimental condition in the present study is provided as a supplementary material.

Various correlations between s_n and ε have been proposed in literature which enable the evaluation of the unstretched flame speed and the burned gas Markstein length by extrapolating s_n to zero stretch rate [58–61]. As the equivalence ratio and the ammonia concentration in the fuel increased, the flame speed-stretch rate relationship became more non-linear. However, the non-linearity decreased with an increase in mixture pressure. Figure 4 shows plots of s_n versus ε for mixtures of different equivalence ratios and ammonia concentrations.

The following non-linear correlation was found appropriate for extracting the unstretched laminar burning velocity from the measured data [59].

$$\left(\frac{S_n}{S_s}\right)^2 \ln\left(\frac{S_n}{S_s}\right)^2 = -2\frac{L_b\varepsilon}{S_s} \tag{3}$$

It can be shown from Eq. (3) that $\ln(s_n)$ varies linearly with $\varepsilon/{s_n}^2$, with an intercept on the $\ln(s_n)$ axis equal to $\ln(s_s)$ and a slope equal to $-s_sL_b$. Hence, s_s and L_b can be obtained from linear extrapolation of $\ln(s_n)$ versus $\varepsilon/{s_n}^2$ plots [60]. The curves in Fig. 4 are the non-linear extrapolations. The flame radii used for the extrapolation ranged from 10 mm (initial radius) to 20~28 mm (final radius) with at least 25 data points as shown in Fig. 4. The unstretched laminar burning velocity, s_l is then obtained by a mass balance across the unstretched flame, which results to $s_l = s_s/\sigma$. Here σ is the ratio of the unburnt mixture density, ρ_u, to the burnt gas density, ρ_b.

3. Numerical simulations

The one-dimensional simulations in this study employed the detailed reaction mechanisms by Glarborg et al. [10], Okafor et al.

Table 2
Reactions updated in or added to the reduced reaction mechanism. Rate coefficient is given by $k_{Ri} = AT^n \exp(-Ea/RT)$. Units are in cm, mol, s, cal and K.

Index	Reaction	A	n	Ea	Source
R3	$O+CH_3=H+CH_2O$	5.540E+13	0.050	−136.00	[68]
R5	$O+CH_2O=OH+HCO$	6.260E+09	1.150	2260.00	[69]
R18	$CH_3+CH_3=C_2H_5+H$	3.160E+13	0.000	14,698.85	[70]
R19	$H+CH_3(+M)=CH_4(+M)$	1.270E+16	−0.630	382.89	[71]
	H_2/2/ H_2O/6/ CO/1.5/ CO_2/2/ CH_4/2/ AR/0.7/				
	LOW / 2.470e+33 −4.760 2440.01 /				
	TROE/ 0.783 74 2941 6964 /				
R21	$H+HCO<=>H_2+CO$	1.210E+14	0.000	0.00	[72]
R36	$OH+HO_2<=>H_2O+O_2$	7.000E+12	0.000	−1092.96	[73]
R37	$OH+HO_2<=>H_2O+O_2$	4.500E+14	0.000	10,929.60	[73]
R43	$CO+OH<=>CO_2+H$	7.015E+04	2.053	−355.70	[74]
R44	$CO+OH<=>CO_2+H$	5.757E+12	−0.664	331.80	[74]
R95	$NH+NO=N_2O+H$	1.800E+14	−0.351	−244.00	[33]
R117	$NH_3+M=NH_2+H+M$	6.600E+17	0.000	97,300.00	[75]

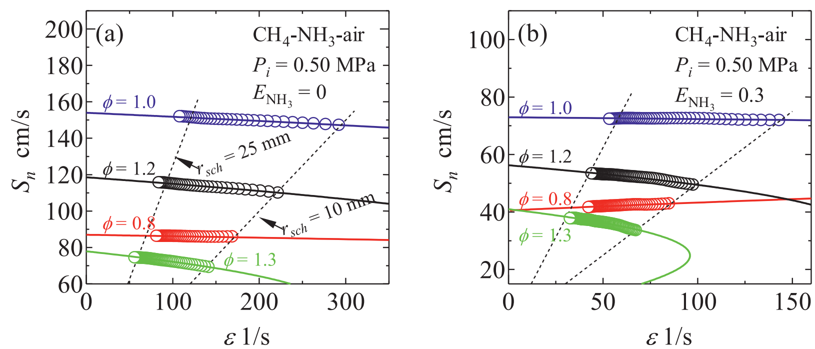

Fig. 4. Plots of s_n versus ε for different equivalence ratios and ammonia concentrations. The symbols represent measured data while the curve are non-linear extrapolation of the measurements.

[21], Tian et al. [28], Mathieu and Petersen [31], and GRI Mech 3.0 [36]. A reduced reaction mechanism with 42 species and 130 reactions was developed from the detailed mechanism by Okafor et al. [21] as described in Section 4 and optimized with the present measurements and data from literature.

Numerical tools and combustion models in ANSYS Chemkin-PRO software version 17.2 [62] were employed in analyzing the oxidation chemistry of the mixtures and in performing one-dimensional simulations. The Flame-Speed Calculation Model was used to calculate the burning velocity of adiabatic and unstretched laminar flames of CH_4–NH_3-air and NH_3-air mixtures for comparison with the present measurements and data in literature. The computation domain size was 10 cm and the unstretched laminar burning velocity was obtained as the velocity at the cold boundary of the computational domain. An automatic estimation of the temperature profile was employed in the calculations involving CH_4–NH_3-air mixtures. On the other hand, a user-specified estimate of the temperature profile was used in the case of NH_3-air mixtures to enhance convergence of the calculations. Furthermore, the Plug Flow Reactor Model was used to model measured NH_3, NO and CO species from CH_4–NH_3–O_2–N_2 oxidation in the temperature-controlled laminar flow reactor experiments by Mendiara and Glarborg. [63]. The experiments were conducted in a 120 cm long alumina reactor with a diameter of 1.2 cm and a residence time in the isothermal zone of 1296/T(K). The simulation applied the experimental temperature profile throughout the whole length of the flow reactor. In addition, the Stagnation Flame Model was used to model the experimental measurements of OH, NO, and NH in strain-stabilized NH_3-air flames by Brackmann et al. [64] employing the experimental temperature profile as an input parameter.

4. Development of the reduced mechanism

Okafor's mechanism [21] was systematically reduced and then optimized against present measurements and literature data. The first step in the reduction process was the application of the Direct Relation Graph with Error Propagation (DRGEP), which identifies unimportant species in a reaction mechanism by resolving species coupling without any priori knowledge of the system. The DRGEP shares the same underlying principle as the DRG developed by Lu and Law [65] in identifying direct species coupling using an immediate error to the production rate of a species owing to the removal of another species. The laminar burning velocity of unstretched adiabatic premixed flames of CH_4–NH_3-air flames were first calculated using the master mechanism from which the production and consumption rate of a species were evaluated. The result was used to construct the direct relation graph with all the species in the mechanism. A specified error tolerance of 10% for the peak values of the target parameters was then used to remove unimportant species from the mechanism. The target parameters in this study were s_l and the concentrations of NH_3, CH_4, H, OH, NO, O and CO over a wide range of equivalence ratios, ammonia concentrations, mixture temperature and pressure. The strength of the GRPEP over the GRP lies in the error propagation function which considers that the influence of an error introduced by a change in the concentration of a species or a removal of the species is damped as it propagates along the graph to the target, leading to more effective mechanism reduction [66,67].

Nevertheless, DRGEP alone did not result to a sufficient reduction of the master mechanism. In DRG or DRGEP, a group of species whose removal may induce an error greater than the user-specified tolerance is retained in the mechanism even though only a few

Table 3
Measured unstretched laminar burning velocity and the corresponding Markstein length at 0.10 MPa.

ϕ [-]	E_{NH_3} [-]	s_l [cm/s]	Range of data scatter from the mean s_l [cm/s]		L_b [mm]	Range of data scatter from the mean L_b [mm]		Le [−] = $\frac{x_{CH_4}Le_{CH_4}+x_{NH_3}Le_{NH_3}}{x_{CH_4}+x_{NH_3}}$
0.7	0	15.17	0.42	−0.37	0.340	0.070	−0.066	0.96
	0.10	12.42	2.24	−0.94	0.314	0.122	−0.007	0.97
	0.20	9.60	0.22	−0.27	0.404	0.074	−0.055	0.97
	0.30	7.59	0.273	−0.28	0.697	0.096	−0.149	0.97
0.9	0.05	25.30	0.52	−0.49	0.481	0.026	−0.021	0.96
1.0	0.05	28.91	0.29	−0.27	0.685	0.026	−0.027	-

Table 4
Measured unstretched laminar burring velocity and the corresponding Markstein length at 0.30 MPa.

ϕ [-]	E_{NH_3} [-]	s_l [cm/s]	Range of data scatter from the mean s_l [cm/s]		L_b [mm]	Range of data scatter from the mean L_b [mm]		Le [−] = $\frac{x_{CH_4}Le_{CH_4}+x_{NH_3}Le_{NH_3}}{x_{CH_4}+x_{NH_3}}$
0.8	0	15.82	0.17	−0.33	0.120	0.001	−0.001	0.96
	0.10	11.44	0.19	−0.26	0.045	0.008	−0.014	0.97
	0.20	9.16	0.10	−0.07	−0.022	0.043	−0.056	0.97
	0.30	7.68	0.00	−0.01	−0.118	0.006	−0.004	0.97
0.9	0	21.02	0.51	−0.49	0.195	0.013	−0.015	0.95
	0.05	17.48	0.59	−0.29	0.187	0.034	−0.056	0.96
	0.10	15.47	0.27	−0.19	0.136	0.004	−0.005	0.97
	0.20	12.62	0.26	−0.22	0.125	0.031	−0.014	0.97
	0.30	10.35	0.10	−0.09	0.109	0.002	−0.003	0.97
1.0	0	24.62	0.33	−0.28	0.264	0.002	−0.004	-
	0.05	20.10	0.41	−0.37	0.256	0.053	−0.055	-
	0.10	17.79	0.08	−0.09	0.239	0.010	−0.019	-
	0.20	14.46	0.10	−0.07	0.220	0.075	−0.045	-
	0.30	11.89	0.02	−0.04	0.260	0.003	−0.004	-
1.1	0	25.25	0.43	−0.24	0.330	0.009	−0.008	1.10
	0.10	17.52	0.33	−0.29	0.347	0.003	−0.004	1.10
	0.20	14.20	0.10	−0.08	0.425	0.009	−0.005	1.10
	0.30	11.66	0.14	−0.10	0.543	0.009	−0.011	1.10
1.2	0	21.27	0.11	−0.15	0.397	0.003	−0.006	1.10
	0.10	14.44	0.16	−0.08	0.553	0.003	−0.004	1.10
	0.20	11.53	0.19	−0.19	0.677	0.010	−0.007	1.10
	0.30	9.35	0.13	−0.07	0.798	0.024	−0.013	1.10
1.3	0	15.06	0.08	−0.11	0.707	0.014	−0.011	1.10
	0.10	10.09	0.15	−0.26	0.845	0.011	−0.019	1.10
	0.20	8.18	0.05	−0.08	0.966	0.027	−0.029	1.10
	0.30	7.22	0.04	−0.06	1.148	0.034	−0.054	1.10

species within the group may be responsible for the noticeable amount of error, while the rest may be safely removed. The species within such a group that lead to significant amounts of induced error and the ones leading to induced error within the tolerance level cannot be treated separately. Therefore, further reduction of the skeletal mechanism was achieved by conducting a brute-force first-order normalized sensitivities of the target parameters, x_i to the rate constants of each reaction step, k_j, given by $\partial \ln x_i / \partial \ln k_j$. The reaction steps which predominantly control the target parameters for the different target conditions were identified and retained in the mechanism while the other less important steps were removed along with the irrelevant species.

The reduced mechanism was then optimized using an extensive set of measured data of the burning velocity, and species measurements of NH_3-air and CH_4–NH_3-air flames. To identify reactions to be optimized, first-order sensitivity of the target parameters were analyzed using the reduced model at different conditions. Table 2 is a list of reactions that were updated or added to the reduced mechanism in order to ensure a satisfactory reproduction of the experimental data set. The index for each reaction is the same as the reaction number in the reduced mechanism.

Okafor's mechanism [21] employs a temperature independent rate constant for (R3) proposed by Slagle et al. [76] and Seakins et al. [77]. However, the reduced model employs more recent temperature dependent theoretical rate constant by Harding et al. [68], which has remarkably high agreement with experimental measurements. In addition, the more recent high fidelity rate constant of $OH+HO_2{=}H_2O+O_2$ proposed by Hong et al. [73] which is expressed by $k_{R36} + k_{R37}$ was adopted in the reduced model. Furthermore, the rate constant of $OH+CO{=}H+CO_2$ employed in the master mechanism which has a monoexponential dependence on temperature is a fit to the data of Yu et al. [78] over the temperature range from 300 to 3000 K by GRI Mech 3.0. However, the reduced mechanism adopts a more recent data by Joshi and Wong [63] for the temperature range from 120 to 2500 K and pressure lower than $P(\mathrm{bar}) = 9 \times 10{-}17T\ 5.9\ \exp(520/T)$, expressed by the combination of $k_{R43} + k_{R44}$ with different temperature exponents, and thus very well captures the biexponential temperature dependence of the series of available measured rate constant over the temperature range. On the other hand, the rate constant of (R95) proposed by Klippeinstein [33] was adopted in this study to avoid the negative value of the A-factor used in the master mechanism. Reaction (R117), which is not in the detailed reaction mechanism was added to the reduced mechanism to improve the performance of the mechanism at high pressures. The relevance of these updated reactions in the prediction of laminar burning velocity and NO concentration by the reduced mechanism is discussed in Section 5.2. The optimized reduced mechanism is provided as a supplementary materials.

Table 5
Measured unstretched laminar burring velocity and the corresponding Markstein length at 0.50 MPa.

ϕ [-]	E_{NH_3} [-]	s_l [cm/s]	Range of data scatter from the mean s_l [cm/s]		L_b [mm]	Range of data scatter from the mean L_b [mm]		$Le\,[-] = \frac{x_{CH4}Le_{CH4}+x_{NH3}Le_{NH3}}{x_{CH4}+x_{NH3}}$
0.8	0	12.93	0.02	−0.04	0.069	0.020	−0.029	0.96
	0.10	9.26	0.12	−0.21	−0.058	0.053	−0.029	0.97
	0.20	7.45	0.02	−0.03	−0.217	0.171	−0.113	0.97
	0.30	6.16	0.15	−0.07	−0.302	0.015	−0.016	0.97
0.9	0	18.09	0.81	−0.78	0.159	0.038	−0.119	0.95
	0.05	14.41	0.40	−0.42	0.133	0.065	0.043	0.96
	0.10	12.98	0.16	−0.25	0.093	0.110	−0.044	0.97
	0.20	10.34	0.27	−0.16	0.030	0.046	−0.078	0.97
	0.30	8.19	0.02	−0.03	−0.051	0.058	−0.116	0.97
1.0	0	20.67	0.34	−0.43	0.235	0.016	−0.020	-
	0.05	16.51	0.21	−0.28	0.204	0.033	−0.037	-
	0.10	14.86	0.24	−0.18	0.135	0.016	−0.017	-
	0.20	11.81	0.03	−0.06	0.157	0.006	−0.006	-
	0.30	9.60	0.06	−0.06	0.054	0.027	−0.037	-
1.1	0	20.48	0.18	−0.18	0.292	0.012	−0.008	1.10
	0.10	14.21	0.34	−0.27	0.213	0.032	−0.031	1.10
	0.20	11.33	0.11	−0.19	0.218	0.001	−0.001	1.10
	0.30	9.56	0.14	−0.18	0.326	0.038	−0.039	1.10
1.2	0	16.23	0.29	−0.32	0.316	0.026	−0.024	1.10
	0.10	11.43	0.16	−0.17	0.330	0.007	−0.007	1.10
	0.20	9.00	0.29	−0.21	0.441	0.014	−0.008	1.10
	0.30	7.66	0.08	−0.07	0.564	0.008	−0.006	1.10
1.3	0	10.54	0.14	−0.13	0.524	0.016	−0.013	1.10
	0.10	7.81	0.04	−0.05	0.580	0.011	−0.015	1.10
	0.20	6.43	0.06	−0.03	0.683	0.015	−0.010	1.10
	0.30	5.63	0.06	−0.03	0.789	0.008	−0.005	1.10

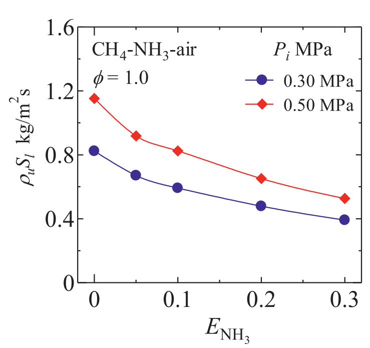

Fig. 5. Variation of the laminar burning flux with hydrogen concentration at 0.30 and 0.50 MPa.

5. Results and discussion

5.1. Measured laminar burning velocity and burned gas Markstein lengths

Mean values of the unstretched laminar burning velocity with the corresponding burned gas Markstein lengths obtained from at least five experimental repetitions at each condition in the present study are presented in Tables 3–5. The range of data scatter from the mean values, i.e. the maximum deviations above and below the mean value in each set of measurement at each condition, is also presented in the table.

The tables shows that the unstretched laminar burning velocity decreased with an increase in ammonia concentration at all mixture pressures. On the other hand, s_l decreased with an increase in mixture pressure. The decreasing trend in s_l with mixture pressure is mainly due to an increase in ρ_u because the laminar burning flux $\rho_u s_l$, which is the eigen value of flame propagation increased with mixture pressure as shown in Fig. 5, indicating an increase in overall reaction rate [79,80].

At 0.30 MPa and 0.50 MPa the Markstein length decreased with an increase in E_{NH_3} for the lean flames, and increased with an increase in E_{NH_3} for the rich flames. Theoretical analysis of premixed flame structure by previous studies shows that the Markstein length, L_b, is a function of the Lewis number, Le, the thermal expansion coefficient, σ, the flame thickness, δ_l, and the Zel'dovich number, Ze. For a weakly stretched flame, these properties may be related thus [60,80,81];

$$L_b = \frac{\delta_l \sigma}{Le} + \delta_l \sigma Ze\left(1 - \frac{1}{Le}\right) \tag{4}$$

The first term in Eq. (4) is always positive and is important even in equidiffusive mixtures. On the other hand, the second term accounts for thermo-diffusive effects on the flame speeds of non-equidiffusive mixtures, and may be negative (positive) for flames with Lewis number less (greater) than unity. It can therefore be inferred that for the lean flames in this study which have $Le < 1$, L_b may decrease with E_{NH_3} if the rate of increase of the absolute value of the second term is more than that of the first term. Furthermore, L_b of the lean flames may become negative, as observed in the measured data, when thermo-diffusive effects becomes dominant.

The variation of the Lewis number, the thermal expansion coefficient, the flame thickness and the Zel'dovich number with ammonia concentration and mixture pressure at $\phi = 0.8$ are shown in Figs. 6–9. The flame thickness, was given by $\delta_l = (T_b - T_u)/(dT/dx)_{max}$ [82], where T_b is the adiabatic flame temperature. On the other hand, the Zel'dovich number was expressed by $Ze = 4(T_b - T_u)/(T_b - T^0)$ [45,80,83], where T^0 is the inner layer temperature.

Figures 10 and 11 show that the first term of Eq. (4) increases with E_{NH_3} for both the lean and rich mixtures, respectively. On the other hand, the second term of the equation is negative (positive) and decreases (increases) with E_{NH_3} for the lean (rich) mixtures. These trends are mainly controlled by the increasing trend in the flame thickness and the Zeldovich number. The calculated rate of decrease of the second term for the lean flame is however not sufficient to result in a decrease in the theoretical L_b with E_{NH_3}. With an increase in pressure, L_b may decrease mainly due to the

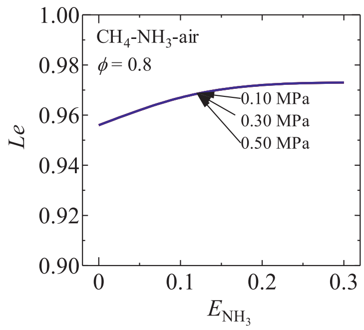

Fig. 6. Variation of *Le* with E_{NH_3} and P_i at $\phi = 0.8$.

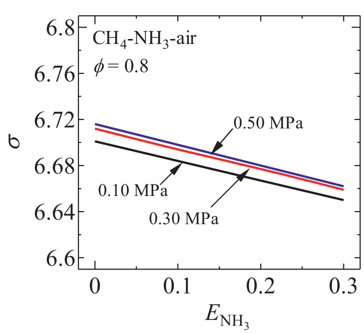

Fig. 7. Variations of σ with E_{NH_3} and P_i at $\phi = 0.8$.

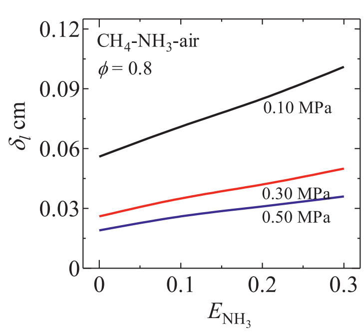

Fig. 8. Variations of δ_l with E_{NH_3} and P_i at $\phi = 0.8$.

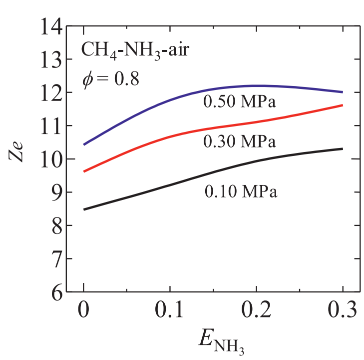

Fig. 9. Variations of *Ze* with E_{NH_3} and P_i at $\phi = 0.8$.

decrease in δ_l as shown in Fig. 8. However, the second term of Eq. (4) is enhanced relative to the first term owing to the increase in the Zel'dovich number. Therefore, the lean flames have a higher tendency of having negative Markstein lengths at higher pressures due to the promotion of thermo-diffusive effects.

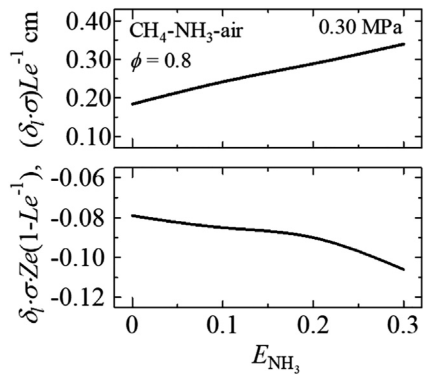

Fig. 10. Trends in the two terms in Eq. (4) at $\phi = 0.8$.

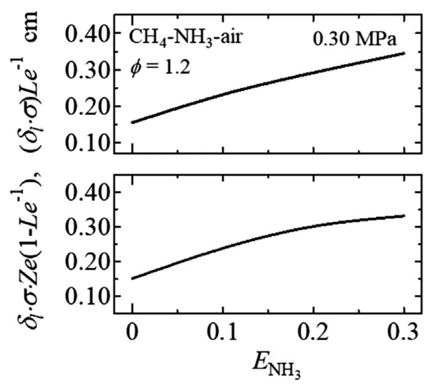

Fig. 11. Trends in the two terms in Eq. (4) at $\phi = 1.2$.

5.2. Validation of the reduced mechanism

5.2.1. Prediction of the unstretched laminar burning velocity

Figure 12 shows a comparison of the measured unstretched laminar burning velocity of stoichiometric CH_4–NH_3-air flames with results simulated using the detailed mechanisms by Okafor et al. [21], Tian et al. [28], GRI Mech 3.0 [36] and the present reduced mechanism. A plot of the percent deviation of the simulated results from the measured values are shown in Fig. 13, with the error bars representing the uncertainty in the measurements in percentage. The reduced mechanism predicts the measurements with high fidelity within the limits of experimental uncertainty. Figure 14 shows that the reduced mechanism satisfactorily reproduces the measured s_l of CH_4–NH_3-air flames over a wide range of conditions of pressure and equivalence ratio. The figure shows that the detailed mechanism by Okafor et al. [21] may predict the burning velocity of the flames at 0.10 MPa satisfactorily, having been optimized at 0.10 MPa. However, the accuracy of its predictions may decrease with an increase in pressure for the rich mixtures of $E_{NH_3} = 0.3$, indicating the need for optimization of the mechanism at high pressures.

First order normalized sensitivities with respect to the unstretched laminar burning velocity were computed using the reduced mechanism to identify the rate limiting reactions in the flame as shown in Fig. 15. The pool of OH, O and H radicals in the flame enhances flame propagation, therefore reactions that control the production or consumption rates of these radicals are the same reactions that have the most influence on s_l. With an increase in ammonia concentration in CH_4–NH_3-air flames, depletion of the O/H radical pool is encouraged by the promotion of chain-terminating reactions from ammonia chemistry which may slow down the overall reaction rate [21]. The reaction $HNO+H = H_2+NO$ and reverse of $NH_3+M = NH_2+H+M$ which

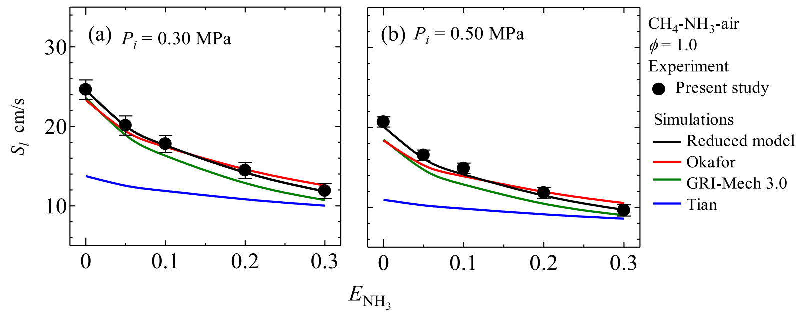

Fig. 12. Comparison of the measured unstretched laminar burning velocity of stoichiometric CH_4–NH_3-air flames from this study to simulated values using the detailed kinetic mechanisms of Okafor et al. [21], Tian et al. [28], GRI Mech 3.0 [36] and the reduced reaction mechanism from this study at (a) 0.30 MPa, (b) 0.50 MPa.

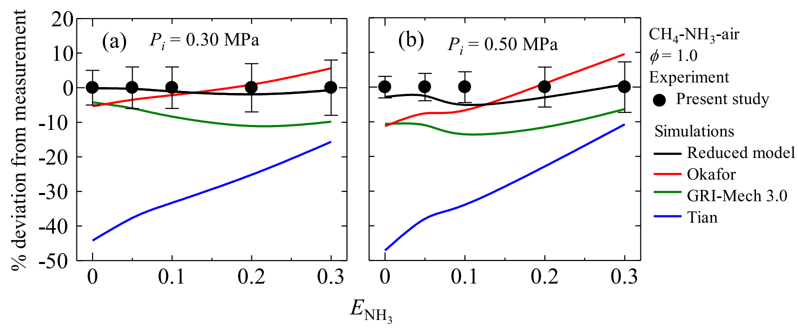

Fig. 13. Deviations of the simulated unstretched laminar burning velocity from the measurements for stoichiometric CH_4–NH_3-air flames at (a) 0.30 MPa, (b) 0.50 MPa. The error bars indicate the uncertainties in the measurements.

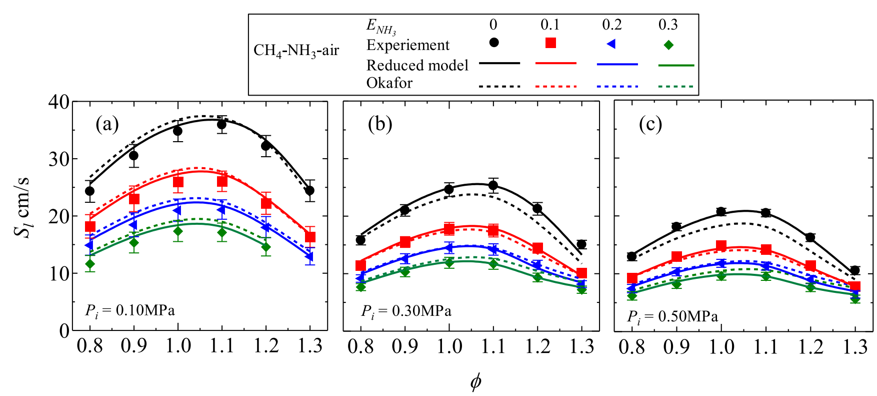

Fig. 14. Comparison of measured unstretched laminar burning velocity of CH_4–NH_3-air flames to simulated values at (a) 0.10 MPa (b) 0.30 MPa (c) 0.50 MPa. The measurements at 0.10 MPa are data from Okafor et al. [21].

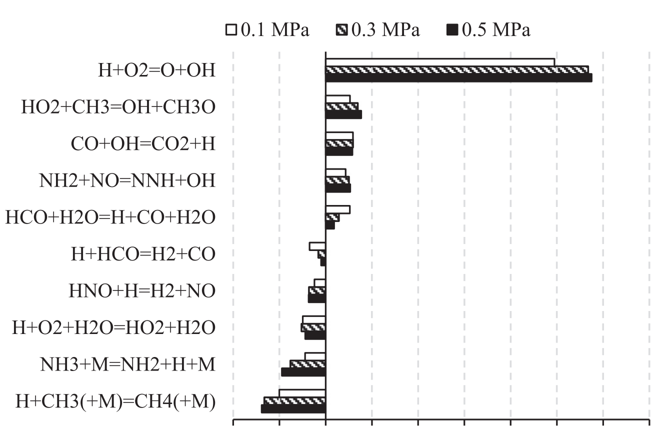

Fig. 15. First order normalized sensitivity of the unstretched laminar burning velocity of stoichiometric CH_4–NH_3-air flames of E_{NH_3} = 0.2.

are major chain-terminating reactions in ammonia oxidation chemistry have significant negative influence on s_l. With an increase in pressure, the O/H radical pool may be depleted due to the enhancement of three body chain-terminating reactions such as $H+CH_3(+M)=CH_4(+M)$, and the reverse of $NH_3+M=NH_2+H+M$ relative to the bimolecular chain-branching or chain-propagating reactions. Figure 16 is a plot of OH, O and H radicals profiles in unstretched adiabatic CH_4–NH_3-air flames at different mixture pressures, showing decrease in the mole fractions of the radicals with an increase in mixture pressure. However, this effect of the relative enhancement of the three-body chain terminating reactions is not sufficient to be responsible for the decrease in s_l with P_i. As explained in Section 5.1, the decrease in s_l with P_i is due to the increase in the unburned mixture density [79,80].

Figure 17 shows that the reduced reaction mechanism also predicts measured unstretched laminar burning velocity of NH_3-air flames in the literature satisfactorily. In Fig. 18, first order normalized sensitivity coefficients with respect to s_l of NH_3-air flames show the important rate-limiting reactions. The reaction,

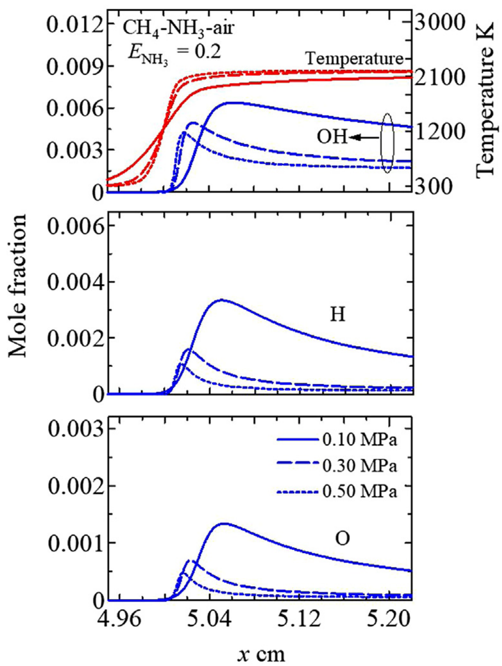

Fig. 16. OH, O and H radicals profiles in stoichiometric unstretched adiabatic CH_4–NH_3-air flames of of E_{NH_3} = 0.2 at different pressures.

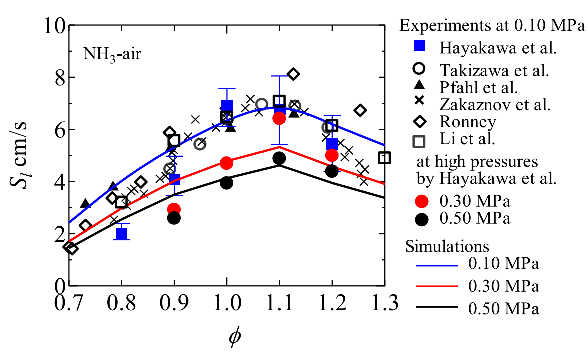

Fig. 17. Comparison of the measured burning velocity of NH_3-air flames by Hayakawa et al. [22], Takizawa et al. [23], Pfahl et al. [24], Zakaznov et al. [25]], Ronney [26] and Li et al. [27] to predictions made by the reduced mechanism.

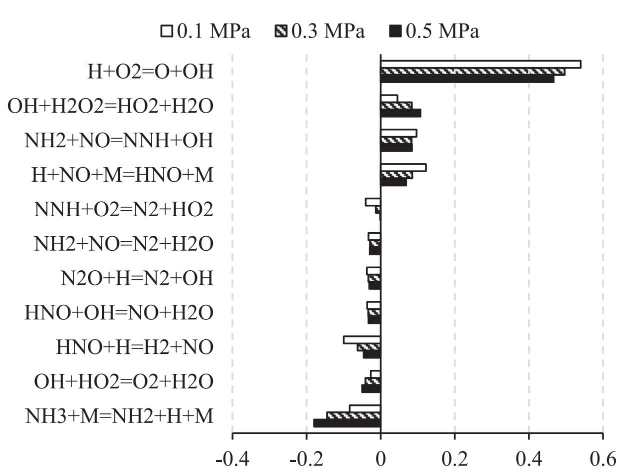

Fig. 18. First order normalized sensitivity of the unstretched laminar burning velocity of stoichiometric NH_3-air flames.

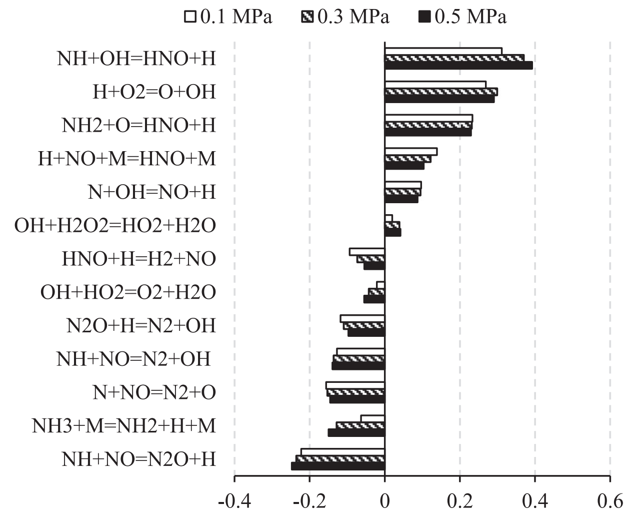

Fig. 19. First order normalized sensitivity of NO concentration at the position of the peak NO concentration in stoichiometric NH_3-air flames.

NH_3+M=NH_2+H + M can be seen to have the most negative influence on s_l owing to the chain-terminating influence of the backward step. The two product channels of NH_2+NO have opposing effects on the burning velocity. The NNH+OH channel is a chain branching step which promotes the production of OH radicals while the N_2+H_2O channel has a chain-terminating character because it competes with the NNH+OH channel. Therefore, the unstretched laminar burning velocity may be sensitive to the branching ratio of the reactions. The branching ratio is also relevant to the Thermal DeNOx process owing to its control of the O/H radicals' production [4,10,32,84–89].

Kobayashi et al. [4] suggested that the accuracy of the prediction of O/H radicals' concentration in NH_3-air flames by a reaction mechanism may correlate with its accuracy to predict the unstretched laminar burning velocity as well as NO concentration. It was found in this study that the set of major rate limiting reactions in NH_3-air flames shown in Fig. 18 are similar to the set of reactions that control NO concentration as shown in Fig. 19. The chain-terminating reactions, which deplete the O/H radical pool exert negative influence on NO concentration while the chain-branching and chain-propagating reactions have positive influence on NO concentration. The reaction HNO+H = H_2+NO is a major NO formation step, however it exerts a negative influence on NO concentration because it is an important chain-terminating reaction in ammonia flames.

Fuel NO formation/reduction is dependent on the O/H radical pool. Glarborg et al. [10] noted that the preference to form NO or N_2 in ammonia oxidation is determined by the competition between the reaction of amine radicals with NO or with the O/H radical pool. The oxidation of NH_i (i = 0, 1, 2, 3) by the O/H radical pool results in NO production predominantly via the HNO intermediate. The abundance of O and OH radicals promotes the conversion of NH_2 and NH to HNO via NH_2+O=HNO+H and NH+OH=HNO+H, which are the predominant HNO formation steps. HNO is then solely converted to NO through reactions with the O/H radicals and through a dissociation reaction. Since HNO contributes to about 70% of NO production from ammonia oxidation [32], the concentration of NO in ammonia-containing flames strongly correlates to the O and OH radical concentrations which are critical to HNO production. Figure 20 shows plots of the maximum NO concentration against the maximum OH, O and H radicals in CH_4–NH_3 -air flames calculated using the reduced mechanism. Plotted in the figures are data for equivalence ratios ranging from 0.7 to 1.3, pressures of 0.1,

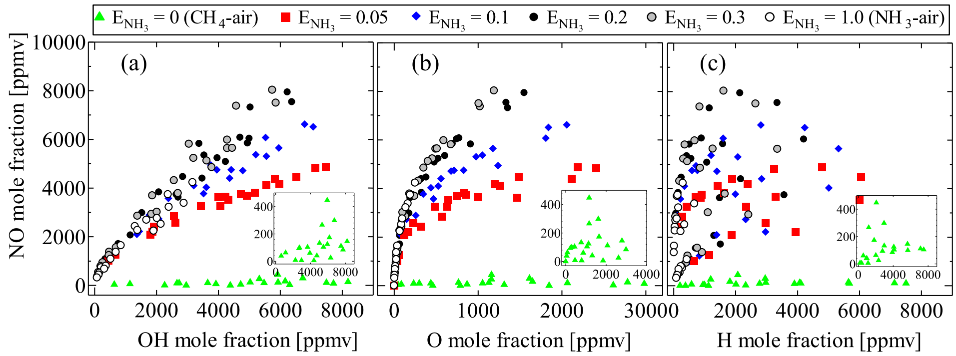

Fig. 20. Plots of the maximum NO concentration against the maximum concentrations of (a) OH, (b) O and (c) H in CH_4–NH_3 -air flames calculated using the reduced mechanism. The inset shows the degree of the scatter for the CH_4-air flames.

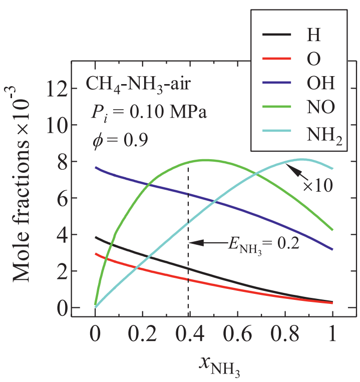

Fig. 21. Variations of the mole fractions of H, O, OH, NO and NH_2 in CH_4–NH_3 -air flames with ammonia mole fraction in the binary fuel.

03, and 0.50 MPa and ammonia heat fraction ranging from 0 to 1.0. For the CH_4-air flame (E_{NH_3} = 0) in which thermal and prompt NO formation routes are dominant, NO concentration does not correlate with the O/H radical concentration as shown by the insets in the figures. With the addition of ammonia to methane, the fuel NOx chemistry becomes the dominant path for NO formation and hence the concentration of NO correlates with those of O and OH.

The correlation however is dependent on ammonia concentration in the binary fuel for $E_{NH_3} < 0.2$. For very low ammonia concentration, NO formation is less sensitive to the concentration of O and OH radicals because even though the concentration of the radicals is high, NO production is limit by the low concentration of NH_i radicals as shown in Fig. 21. The sensitivity of the concentration of NO to those of O and OH radicals increases with E_{NH_3} as NH_i radicals concentration increases and tends to saturate for high ammonia concentration of $E_{NH_3} > 0.2$. The trend in NO concentration reflects the decreasing trend in the concentration of O/H radicals only for high ammonia concentrations as shown in Fig. 21.

On the other hand, the correlation of NO with H is dependent on equivalence ratio, hence the plots in Fig. 20(c) do not collapse. Depending on the equivalence ratio, the contribution of H radicals to NO formation/reduction varies. The oxidation of NH_2 and NH by H radicals is promoted as the flame gets richer, leading to the production of N atoms which enhance NO reduction through $N+NO=N_2+O$ [4].

From the foregoing, it can be concluded that the depletion of the O/H radical pool with an increase in pressure may lead to a decrease in fuel NO formation. Experimental studies have reported a decrease in NOx emission from NH_3-air flames in swirl combustors with an increase in pressure [2,4]. In addition, combustion of

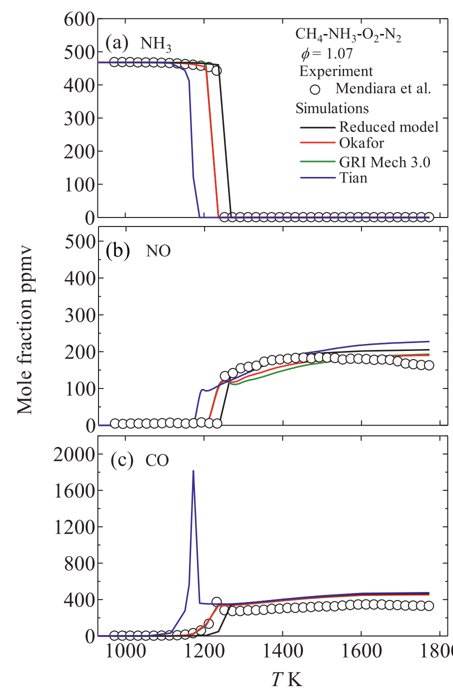

Fig. 22. Comparison of experimental data [63] with modelling predictions for oxidation of a stoichiometric N_2 diluted CH_4–NH_3–O_2 mixture as a function of temperature. Inlet concentrations CH_4 = 2513 ppm, O_2 = 5040 ppm, NH_3 = 468 ppm; balance N_2.

CH_4–NH_3-air mixtures with more than 10% NH_3 volume fraction in the fuel may produce more NO emissions than NH_3-air mixtures as reported by Kurata et al. [1] owing to the larger pool of O/H radicals in CH_4–NH_3-air flames.

5.2.2. Prediction of species concentration

The measurements by Mendiara and Glarborg [63] of NH_3, NO, and CO concentrations at the exit of a temperature controlled laminar flow reactor supplied with a N_2 diluted CH_4–NH_3–O_2 mixtures were modeled as described in Section 3. Figure 22 shows a comparison of the measurements with numerical results at stoichiometric condition. The figure shows that the reduced mechanism models the consumption of NH_3, and the production of NO, and CO in the experiment satisfactorily.

NO formation/consumption in the laminar flow reactor was found to be primarily through the fuel NOx chemistry as shown in Fig. 23 for reactor temperatures of 1270 and 1600 K. The HNO route alone accounts for about 70% of total NO formation and about 54%

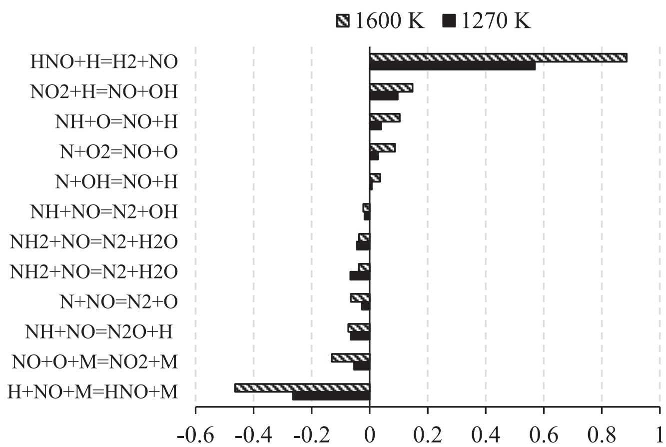

Fig. 23. Integrated rate of production of NO in the laminar flow reactor for the stoichiometric mixture normalize by the integrated absolute rate of NH_3 consumption from NH_3+OH=NH_2+H_2O calculated using the reduced mechanism.

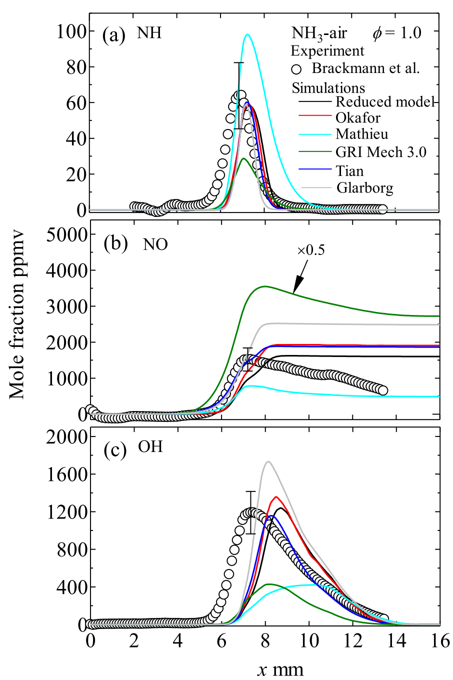

Fig. 24. Comparison of experimental measurements [64] of species concentration in strain-stabilized stoichiometric NH_3-air flames with modelling predictions.

of total NO consumption at 1600 K. On the other hand, the 3-step extended Zel'dovich mechanism contributes to about 9% and 13% of total NO formation and consumption respectively at 1600 K. The first step of the mechanism, N_2+O = N+NO, is an NO reduction step because the abundance of N atoms from the oxidation of NH_i radicals shifts the equilibrium of the reaction to the left, thereby promoting the NO-reducing backward step.

Furthermore, measured OH, NH and NO profiles in the strain-stabilized NH_3-air flames of Brackmann et al. [64] were modeled as shown in Fig. 24. The reduced reaction mechanism and the detailed reaction mechanisms of Okafor et al. [21] and Tian et al. [28] satisfactorily predict the peak values of the species profiles. The measured NO profile exhibits a decreasing trend in the post flame zone in contrast to the trend in the simulations. The experimental evaluation of the species profiles was from LIF images hence the decreasing NO concentration in the post flame was attributed to a possible laser signal absorption. Furthermore, a shift of the predicted profile downstream of the flame in comparison to the measured profile can be observed. This shift was attributed to a possible offset between the temperature profile which was measured with CARS and the species profiles which were evaluated using LIF images [64]. Note that the simulations here employed the experimental temperature profiles.

For the mixture of Fig. 22 with E_{NH_3} = 0.07, GRI Mech 3.0 closely models the NO formation. However, the detailed mechanism may over-predict NO emission for mixtures with higher ammonia concentrations [21,38] owing to its significant over-prediction of NO production from ammonia oxidation as suggested in Fig. 24. The absence of the NH_2+NO reactions in GRI Mech 3.0 contributes to the over-prediction of NO concentration [21]. On the other hand, the radical combination reactions NH_i+NH_i which lead to the production of N_2H_i species and then N_2 without significant NO production may contribute to lower NO production in ammonia flames, especially in rich flame conditions [90,91]. Again, these reactions are not included in GRI Mech 3.0. It is worth noting however that the N_2H_i chemistry subset is also omitted in Mathieu's mechanism [31]. Mathieu and Petersen [31] found out that the effect of the N_2H_i chemistry subset is the reason the detailed mechanism by Klippeinstein et al. [33] does not satisfactory model their shock tube measurements of the ignition delay times of ammonia flames. It has been suggested by Kobayashi et al. [4] and Glarborg et al. [10] that this chemistry subset has not been well characterized, however its importance in ammonia chemistry has been variously reported [64,90–92]. The reaction NH_2+NH=N_2H_2+H is the most important single reaction step controlling the concentration of NH radicals in stoichiometric and rich ammonia flames [64,92], contributing to about 30–35% of total NH consumption in stoichiometric flames according to the detailed reaction mechanisms of Glarborg et al. [10], Okafor et al.[21], Tian et al. [28] and the present reduced model.

From the discussion in Section 5.2.1, it can be deduced that reaction mechanisms that underestimate or overestimate OH concentration in ammonia flames may also underestimate or overestimate, respectively the NO concentration as seen in Fig. 24 for the case of Mathieu's [31] and Glarborg's mechanisms [10]. However, it was found that GRI Mech 3.0 is an exception mostly due to the significant dependence of HNO and, thus, NO formation on NH+H_2O=HNO+H_2 in the chemistry. According to GRI Mech 3.0, the reaction is the most important HNO production step and a major NH consumption step in NH_3-air flames, and contributes predominantly to the high NO concentration predicted by the mechanism. However, studies have shown that the reaction NH+H_2O=HNO+H_2 is not important in fuel NOx chemistry. GRI Mech 3.0 adopts the rate constants by Rohrig and Wagner [93] who experimentally studied different product channels for NH+H_2O. They proposed that the reactions are fast and that the most likely product channel is HNO+H_2. A later study by Glarborg et al. [94] found that the rate proposed by Rohrig and Wagner [93] for NH+H_2O=HNO+H_2 is too fast to model their experimental measurement of NO reduction by H_2 in a fast-mixing flow reactor. Hence they omitted the reaction from their mechanism. The study of the effects of NO and SO_2 on the oxidation of a CO–H_2 mixture in a jet-stirred reactor by Dagaut et al. [95] also ruled out the possibility of any significant HNO to NH conversion through the reverse reaction. Fontijn et al. [96] reinvestigated the product channels of NH+H_2O reaction and concluded that the most likely products of the reaction are not HNO+H_2, and the study by Mackie and Bacskay [97] showed that the product of the reaction is rather NH_2+OH. Shmakov et al. [35] reported that the replacement of the reaction NH+H_2O=HNO+H_2 with NH+H_2O=NH_2OH in Konnov's mechanism improved the agreement between measured and simulated NO and NH_3 profiles in H_2+O_2+N_2 flames doped with either NO or NH_3.

6. Conclusions

This study reports an extensive set of measurements of the unstretched laminar burning velocity and Markstein lengths of CH_4–NH_3-air flames at high pressures up to 0.50 MPa using a constant volume chamber. The unstretched laminar burning velocity decreased with an increase in ammonia concentration in the fuel and mixture pressure. It was found that at high pressures the Markstein length decreased with an increase in E_{NH_3} for the lean flames, and increased with an increase in E_{NH_3} for the rich flames. A reduced reaction mechanism was developed and optimized with the present measurements and data in literature. It is shown that the reduced mechanism models the unstretched laminar burning velocity of and species concentration in NH_3-air and CH_4–NH_3-air flames with high fidelity. Furthermore, it was found that the rate limiting reactions in NH_3-air flames are important reaction steps controlling the NO concentration in the flame owing to the strong dependence of fuel NO chemistry on the O/H radical pool. Therefore, accurate prediction of the laminar burning velocity and NO concentration in ammonia containing flames may strongly depend on the accuracy of the prediction of the O/H radical's concentration. Combustion of CH_4–NH_3-air mixtures with more that 10% NH_3 volume fraction in the fuel may produce more NO emissions than NH_3-air mixtures owing to the larger pool of O/H radicals in the CH_4–NH_3-air flames. The prediction of NO formation in the oxidation of mixtures with high ammonia concentration by GRI Mech 3.0 was found to be predominantly influenced by the reaction $NH+H_2O=HNO+H_2$ which in fact may not be important in ammonia oxidation or the fuel NO chemistry.

Acknowledgments

This study was supported by the Council for Science, Technology and Innovation (CSTI), the Cross-ministerial Strategic Innovation Promotion Program (SIP) "Energy Carriers" (Funding agency: Japan Science and Technology Agency (JST)).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2019.03.008.

References

[1] O. Kurata, N. Iki, T. Matsunuma, T. Inoue, T. Tsujimura, H. Furutani, H. Kobayashi, A. Hayakawa, Performances and emission characteristics of NH_3-air and NH_3-CH_4-air combustion gas-turbine power generations, Proc. Combust. Inst. 36 (2017) 3351–3359.
[2] E.C. Okafor, K.D.K.A Somarathne, A. Hayakawa, T. Kudo, O. Kurata, N. Iki, H. Kobayashi, Towards the development of an efficient low-NOx ammonia combustor for a micro gas turbine, Proc. Combust. Inst. 37 (2019) 4597–4606.
[3] O. Kurata, N. Iki, T. Inoue, T. Matsunuma, T. Tsujimura, H. Furutani, M Kawano, K. Arai, E.C. Okafor, A. Hayakawa, H. Kobayashi, Development of a wide range–operable, rich-lean low-NOx combustor for NH_3 fuel gas-turbine power generation, Proc. Combust. Inst. 37 (2019) 4589–4595.
[4] H. Kobayashi, A. Hayakawa, K.D.K.A Somarathne, E.C. Okafor, Science and Technology of ammonia combustion, Proc. Combust. Inst. 37 (2019) 109–133.
[5] A. Valera-Medina, H. Xiao, M. Owen-Jones, W.I.F. David, P.J. Bowen, Ammonia for power, Prog. Energy Combust. Sci. 69 (2018) 63–102.
[6] D.T. Pratt, Performance of ammonia-fired gas- turbine combustors, Berkley University of Cali- fornia, 1967 Technical Report No.9, DA- 04-200-AMC-791(x)available at http://www.dtic.mil/dtic/tr/fulltext/u2/657585.pdf.
[7] N. Cohen, The $O+NH_3$ reaction: a review, Int. J. Chem. Kinet. 19 (1987) 319–362.
[8] J.A. Miller, C.T. Bowman, Mechanism and modelling of nitrogen chemistry in combustion, Prog. Energy Combust. Sci. 15 (1989) 287–338.
[9] A.M. Dean AM, J.W. Bozzelli, Combustion chemistry of nitrogen, in: W.C. Gardiner Jr. (Ed.), Gas phase combustion chemistry, Springer, New York, 2000, pp. 125–341.
[10] P. Glarborg, J.A. Miller, B. Ruscic, S.J. Klippenstein, Modelling nitrogen chemistry in combustion, Prog. Energy Combust. Sci. 67 (2018) 31–68.
[11] K.D.K.A. Somarathne, A. Hayakawa, H. Kobayashi, Numerical investigation on the combustion characteristics of turbulent premixed ammonia/air flames stabilized by a swirl burner, J. Fluid Sci. Technol. 11 (2016) Paper No. 16-00126, https://doi.org/10.1299/jfst.2016jfst0026.
[12] K.D.K.A. Somarathne, S. Hatakeyama, A. Hayakawa, H. Kobayashi, Numerical study of a low emission gas turbine like combustor for turbulent ammonia/air premixed swirl flames with a secondary air injection at high pressure, Int. J. Hydrog. Energy 42 (2017) 27388–27399.
[13] K.D.K.A. Somarathne, S. Colson, A. Hayakawa, H. Kobayashi, Modelling of ammonia/air non-premixed turbulent swirling flames in a gas turbine-like combustor at various pressures, Combust. Theory Model. 22 (2018) 973–997.
[14] P. Kumar, T.R. Meyer, Experimental and modeling study of chemical-kinetics mechanisms for H_2-NH_3-air mixtures in laminar premixed jet flames, Fuel 108 (2013) 166–176.
[15] J. Li, H. Huang, N. Kobayashi, Z. He, Y. Nagai, Study on using hydrogen and ammonia as fuels: combustion characteristics and NOx formation, Int. J. Energy Res. 38 (2014) 1214–1223.
[16] A. Ichikawa, A. Hayakawa, Y. Kitagawa, K.D.K.A. Somarathne, T. Kudo, H. Kobayashi, Laminar burning velocity and Markstein length of ammonia/hydrogen/air premixed flames at elevated pressures, Int. J. Hydrog. Energy 40 (2015) 9570–9578.
[17] D. Pugh, P. Bowen, A. Valera-Medina, A. Giles, J. Runyon, R. Marsh, Influence of steam addition and elevated ambient conditions on NOx reduction in a staged premixed swirling NH_3/H_2 flame, Proc. Combust. Inst. 37 (2019) 5401–5409.
[18] A.J. Reiter, S.C. Kong, Combustion and emissions characteristics of compression-ignition engine using dual ammonia-diesel fuel, Fuel 90 (2011) 87–97.
[19] A.J. Reiter, S.C. Kong, Demonstration of compression-ignition engine combustion using ammonia in reducing greenhouse gas emissions, Energy Fuels 22 (2008) 2963–2971.
[20] A. Valera-Medina, S. Morris, J. Runyon, D.G. Pugh, R. Marsh, P. Beasley, T. Hughes, Ammonia, methane and hydrogen for gas turbines, Energy Proc. 75 (2015) 118–123.
[21] E.C. Okafor, Y. Naito, S. Colson, A. Ichikawa, T. Kudo, A. Hayakawa, H. Kobayashi, Experimental and numerical study of the laminar burning velocity of CH_4–NH_3–air premixed flames, Combust. Flame 187 (2018) 185–198.
[22] A. Hayakawa, T. Goto, R. Mimoto, Y. Arakawa, T. Kudo, H. Kobayashi, Laminar burning velocity and Markstein length of ammonia/air premixed flames at various pressures, Fuel 159 (2015) 98–106.
[23] K. Takizawa, A. Takahashi, K. Tokuhashi, S. Kondo, A. Sekiya, Burning velocity measurements of nitrogen-containing compounds, J. Hazard. Mater. 155 (2008) 144–152.
[24] U.J. Pfahl, M.C. Ross, J.E. Shepherd, K.O. Pasamehmetoglu, C. Unal, Flammability limits, ignition energy, and flame speeds in H2- CH_4-NH_3-N_2O-O_2-N_2 mixtures, Combust. Flame 123 (2000) 140–158.
[25] F.Z. Zakaznov, L.A. Kursheva, Z.I. Felina, Determination of normal flame velocity and critical diameter of flame extinction in ammonia-air mixture, Combust. Explos. Shock Waves 14 (1978) 710–713.
[26] P.D. Ronney, Effect of chemistry and transport properties on near-limit flames at microgravity, Combust. Sci. Technol. 59 (1988) 123–141.
[27] Y. Li, M. Bi, B. Li, W. Gao, Explosion behaviours of ammonia-air mixtures, Combust. Sci. Technol. 190 (2018) 1804–1816.
[28] Z. Tian, L. Zhang, Y. Li, T. Yuan, F. Qi, An experimental and kinetic modelling study of a premixed nitromethane flame at low pressure, Proc. Combust. Inst. 32 (I) (2009) 311–318.
[30] J.A. Miller, M.D. Smooke, R.M. Green, R.J. Kee, Kinetic modelling of the oxidation of ammonia in flames, Combust. Sci. Technol. 34 (1983) 149–176.
[31] O. Mathieu, E.L. Petersen, Experimental and modelling study on the high-temperature oxidation of Ammonia and related NOx chemistry, Combust. Flame 162 (2015) 554–570.
[32] R.P. Lindstedt, F.C. Lockwood, M.A. Selim, Detailed kinetic modelling of chemistry and temperature effects on ammonia oxidation, Combust. Sci. Technol. 99 (1994) 253–276.
[33] S.J. Klippenstein, L.B. Harding, P. Glarborg, J.A. Miller, The role of NNH in NO formation and control, Combust. Flame 158 (2011) 774–789.
[34] P. Dagaut, P. Glarborg, M.U. Alzueta, The oxidation of hydrogen cyanide and related chemistry, Prog. Energy Combust. Sci. 34 (2008) 1–46.
[35] A.G. Shmakov O. P. Korobeinichev, I.V. Rybitskaya, A.A. Chernov, D.A. Knyazkov, T.A. Bolshova, A.A. Konnov, Formation and consumption of NO in $H_2+O_2+N_2$ flames doped with NO or NH_3 at atmospheric pressure, Combust. Flame 157 (2010) 556–565.
[36] G.P. Smith, D.M. Golden, M. Frenklach, N.W. Moriarty, B. Eiteneer, M. Goldenberg, et al., GRI Mech 3.0. gas research institute, Available at http://www.me.berkeley.edu/gri_mech/.
[37] H. Nakamura, S. Hasegawa, T. Tezuka, Kinetic modelling of ammonia/air weak flames in a micro flow reactor with a controlled temperature profile, Combust. Flame 185 (2017) 16–27.
[38] H. Xiao, A. Valera-Medina, R. Marsh, P.J. Bowen, Numerical study assessing various ammonia/methane reaction models for use under gas turbine conditions, Fuel 196 (2017) 344–351.
[39] S. Hochgreb, Mind the gap: turbulent combustion model validation and future needs, Proc. Combust. Inst. 37 (2019) 2091–2107.
[40] T. Lu, C.K. Law, Toward accommodating realistic fuel chemistry in large-scale computations, Prog. Energy Combust. Sci. 35 (2009) 192–215.
[41] D. Bradley, M. Lawes, M.S. Mansour, Correlation of turbulent burning velocities of ethanol–air, measured in a fan-stirred bomb up to 1.2 MPa, Combust. Flame 158 (2011) 123–138.

[42] D. Bradley, A.K.U. Lau, M. Lawes, Flames stretch rate as a determinant of turbulent burning velocity, Philos. Trans. R. Soc. Lond. A 338 (1992) 359–387.
[43] T. Kitagawa, T. Nakahara, K. Maruyama, K. Kado, A. Hayakawa, S. Kobayashi, Turbulent burning velocity of hydrogen-air premixed propagating flames at elevated pressures, Int. J. Hydrog. Energy 33 (2008) 5842–5849.
[44] E.C. Okafor, A. Hayakawa, Y. Nagano, T. Kitagawa, Effects of hydrogen concentration on premixed laminar flames of hydrogen-methane-air, Int. J. Hydrog. Energy 39 (2014) 2409–2417.
[45] E.C. Okafor, Y. Nagano, T. Kitagawa, Experimental and theoretical analysis of cellular instability in lean H_2-CH_4-air flames at elevated pressures, Int. J. Hydrog. Energy 41 (2016) 6581–6592.
[46] F.N. Egolfopoulos, N. Hansen, Y. Ju, K. Kohse-Höinghaus, C.K. Law, F. Qi, Advances and challenges in laminar flame experiments and implications for combustion chemistry, Prog. Energy Combust. Sci. 43 (2014) 36–67.
[47] Z. Chen, On the accuracy of laminar flame speeds measured from outwardly propagating spherical flames: methane/air at normal temperature and pressure, Combust. Flame 162 (2015) 2442–2453.
[48] A.A. Konnov, A. Mohammad, V.R. Kishore, N. Il Kim, C. Prathap, S. Kumar, A comprehensive review of measurements and data analysis of laminar burning velocities for various fuel+air mixtures, Prog. Energy Combust. Sci. 68 (2018) 197–267.
[49] H. Yu, W. Han, J. Santner, X. Gou, C.H. Sohn, Y. Ju, Z. Chen, Radiation-induced uncertainty in laminar flame speed measured from propagating spherical flames, Combust. Flame 161 (2014) 2815–2824.
[50] J. Jayachandran, R. Zhao, F.N. Egolfopoulos, Determination of laminar flame speeds using stagnation and spherically expanding flames: molecular transport and radiation effects, Combust. Flame 161 (2014) 2305–2316.
[51] G. Rozenchan, D.L. Zhu, C.K. Law, S.D. Tse, Outward propagation, burning velocities, and chemical effects of methane flames up to 60 ATM, Proc. Combust. Inst. 29 (2002) 1461–1470.
[52] D. Bradley, C.M. Harper, The development of instabilities in laminar explosion flames, Combust. Flame 99 (1994) 562–572.
[53] D. Bradley, R.A. Hicks, M. Lawes, C.G.W. Sheppard, R. Woolley, The measurement of laminar burning velocities and Markstein numbers for iso-octane –air and iso-octane – n-heptane – air mixtures at elevated temperatures and pressures in an explosion bomb, Combust. Flame 115 (1998) 126–144.
[54] G. Wang, Y. Li, W. Yuan, Y. Wang, Z. Zhou, Y. Liu, J. Cai, Investigation on laminar flame propagation of n-butanol/air and n-butanol/O_2/He mixtures at pressures up to 20 atm, Combust. Flame 191 (2018) 368–380.
[55] P.D. Ronney, H.Y. Wachman, Effect of gravity on laminar premixed gas combustion I: flammability limits and burning velocities, Combust. Flame 62 (1985) 107–119.
[56] J. Huo, S. Yang, Z. Ren, D. Zhu, C.K. Law, Uncertainty reduction in laminar flame speed extrapolation for expanding spherical flames, Combust. Flame 189 (2018) 155–162.
[57] M. Ben Yaala, L. Marot, R. Steiner, L. Moser, G. De Temmerman, C. Porosnicu, C.P. Lungu, M. Oberkofler, E. Meyer, Quartz micro-balance and in situ XPS study of the adsorption and decomposition of ammonia on gold, tungsten, boron, beryllium and stainless steel surfaces, Nucl. Fus. 58 (2018) 106012.
[58] D. Bradley, P.H. Gaskell, X.J. Gu, Burning velocities, Markstein lengths, and flame quenching for spherical methane-air flames: a computational study, Combust. Flame 104 (1996) 176–198.
[59] A.P. Kelley, C.K. Law, Nonlinear effects in the extraction of laminar flame speeds from expanding spherical flames, Combust. Flame 156 (2009) 1844–1851.
[60] Z. Chen, On the extraction of laminar flame speed and Markstein length from outwardly propagating spherical flames, Combust. Flame 158 (2011) 291–300.
[61] F. Wu, W. Liang, Z. Chen, Y. Ju, C.K. Law, Uncertainty in stretch extrapolation of laminar flame speed from expanding spherical flames, Proc. Combust. Inst. 35 (2015) 663–670.
[62] , CHEMKIN-PRO 17.2, ANSYS, Inc., San Diego, 2016.
[63] T. Mendiara, P. Glarborg, Ammonia chemistry in oxy-fuel combustion of methane, Combust. Flame 156 (2009) 1937–1949.
[64] C. Brackmann, V.A. Alekseev, B. Zhou, E. Nordström, P. Bengtsson, Z. Li, M. Aldén, A.A. Konnov, Structure of premixed ammonia+air flames at atmospheric pressure: laser diagnostics and kinetic modelling, Combust. Flame 163 (2016) 370–381.
[65] T. Lu, C.K. Law, A directed relation graph method for mechanism reduction, Proc. Combust. Inst. 30 (2005) 1333–1341.
[66] K.E. Niemeyer, C.J. Sung, M.P. Raju, Skeletal mechanism generation for surrogate fuels using directed relation graph with error propagation and sensitivity analysis, Combust. Flame 157 (2010) 1760–1770.
[67] Y. Chen, J.-Y. Chen, Application of Jacobian defined direct interaction coefficient in DRGEP-based chemical mechanism reduction methods using different graph search algorithms, Combust. Flame 174 (2016) 77–84.
[68] L.B. Harding, S.J. Klippenstein, Y. Georgievskii, Reactions of oxygen atoms with hydrocarbon radicals: a priori kinetic predictions for the CH_3+O, C_2H_5+ O, and C_2H_3+ O reactions, Proc. Combust. Inst. 30 (2005) 985–992.
[69] W.K. Metcalfe, S.M. Burke, S.S. Ahmed, H.J. Curran, A hierarchical and comparative kinetic modelling study of C_1 - C_2 hydrocarbon and oxygenated fuels, Int. J. Chem. Kinet. 45 (2013) 638–675.
[70] K.P. Kim, J.V. Michael, The thermal reactions of CH_3, Symp. (Int.) Combust. 25 (1994) 713–719.
[71] C.T. Bowman, R.K. Hanson, D.F. Davidson, W.C. Gardiner, Jr., V. Lissianski, G.P. Smith, D.M. Golden, M. Frenklach and M. Goldenberg, http://www.me.berkeley.edu/gri_mech/
[72] W. Tsang, R.F. Hampson, Chemical kinetic data base for combustion chemistry. Part I. Methane and related compounds, J. Phys. Chem. Ref. Data 15 (1986) 1087–1279.
[73] Z. Hong, K.Y. Lam, R. Sur, S. Wang, D.F. Davidson, R.K. Hanson, On the rate constants of $OH+HO_2$ and HO_2+HO_2: a comprehensive study of H_2O_2 thermal decomposition using multi-species laser absorption, Proc. Combust. Inst. 34 (2013) 565–571.
[74] A.V. Joshi, H. Wang, Master equation modelling of wide range temperature and pressure dependence of $CO+OH \rightarrow$ products, Int. J. Chem. Kinet. 38 (2006) 57–73.
[75] B.H. Cardelino, C.E. Moore, C.A. Cardelino, S.D. McCall, D.O. Frazier, K.J. Bachmann, Semiclassical calculation of reaction rate constants for homolytical dissociation reactions of interest in organometallic vapor-phase epitaxy (OMVPE), J. Phys. Chem. A 107 (2003) 3708–3718.
[76] I.R. Slagle, D. Sarzyński, D. Gutman, Kinetics of the reaction between methyl radicals and oxygen atoms between 294 and 900 K, J. Phys. Chem. 91 (1987) 4375–4379.
[77] P.W. Seakins, S.R. Leone, A laser flash photolysis/time-resolved FTIR emission study of a new channel in the reaction of methyl + oxygen atom: production of carbon monoxide(v), J. Phys. Chem. 96 (1992) 4478–4485.
[78] C.- L. Yu, C. Wang, M. Frenklach, Measurement of the rate coefficient of reaction $CO+OH=CO_2+H$, 23rd fall technical meeting of the eastern section of the combustion institute (1990), paper 20.
[79] C.K. Law, Propagation, structure, and limit phenomena of laminar flames at elevated pressures, Combust. Sci. Technol. 178 (2006) 335–360.
[80] C.K. Law, Combustion physics, Cambridge University Press, New York, USA, 2006.
[81] Z. Chen, M.P. Burke, Y. Ju, Effects of Lewis number and ignition energy on the determination of laminar flame speed using propagating spherical flames, Proc. Combust. Inst. 32 (I) (2009) 1253–1260.
[82] C.J. Sun, C.J. Sung, L. He, C.K. Law, Dynamics of weakly stretched flames: quantitative description and extraction of global flame parameters, Combust. Flame 118 (1999) 108–128.
[83] U.C. Müller, M. Bollig, N. Peters, Approximations for burning velocities and Markstein numbers for lean hydrocarbon and methanol flames, Combust. Flame 108 (1997) 349–356.
[84] J.A. Miller, C.T. Bowman, Mechanism and modelling of nitrogen chemistry in combustion, Prog. Energy Combust. Sci. 15 (1989) 287–338.
[85] P. Glarborg, P. Kristensen, K. Dam-Johansen, J.A. Miller, Branching Fraction of the NH_2+NO Reaction between 1210 and 1370 K, J. Phys. Chem. A 101 (1997) 3741–3745.
[86] J.A. Miller, P. Glarborg, Modeling the thermal De-NOx process: closing in on a final solution, Int. J. Chem. Kinet. 31 (1999) 757–765.
[87] P. Glarborg, K. Dam-Johansen, J.A. Miller, R.J. Kee, M.E. Coltrin, Modelling the thermal DeNOx process in flow reactors. Surface effects and nitrous oxide formation, Int. J. Chem. Kinet. 26 (1994) 421–436.
[88] R.P. Lindstedt, F.C. Lockwood, M.A. Selim, A detailed kinetic study of ammonia oxidation, Combust. Sci. Technol. 108 (1995) 231–254.
[89] J.A. Miller, S.J. Klippenstein, Theoretical considerations in the NH_2+ NO reaction, J. Phys. Chem. A 104 (2000) 2061–2069.
[90] A.M. Dean, M. -S Chou, D. Stern, Kinetics of rich ammonia flames, Int. J. Chem. Kinet. 16 (1984) 633–653.
[91] B.S. Haynes, Reactions of ammonia and nitric oxide in the burnt gases of fuel-rich hydrocarbon-air flames, Combust. Flame 28 (1977) 81–91.
[92] A.A. Konnov, J. De Ruyck, Kinetic modeling of the thermal decomposition of ammonia, Combust. Sci. Technol. 152 (2000) 23–37.
[93] M. Rohrig, H.G. Wagner, The reactions of NH ($X^3\Sigma^-$) with the water gas components CO_2, H_2O, and H2, (2000) 975–981.
[94] P. Glarborg, P.G. Kristensen, K. Dam-Johansen, M.U. Alzueta, A. Millera, R. Bilbao, Nitric oxide reduction by non-hydrocarbon fuels. Implications for reburning with gasification gases, Energy Fuels 14 (2000) 828–838.
[95] P. Dagaut, F. Lecomte, J. Mieritz, P. Glarborg, Experimental and kinetic modeling study of the effect of NO and SO_2 on the oxidation of CO+H_2 mixtures, Int. J. Chem. Kinet. 35 (2003) 564–575.
[96] A. Fontijn, S.M. Shamsuddin, D. Crammond, P. Marshall, W.R. Anderson, Kinetics of the NH reaction with H_2 and reassessment of HNO formation from NH^+ CO_2, H_2O, Combust. Flame 145 (2006) 543–551.
[97] J.C. Mackie, G.B. Bacskay, Quantum chemical study of the mechanism of reaction between NH ($X^3\Sigma^-$) and H_2, H_2O, and CO_2 under combustion conditions, J. Phys. Chem. A 109 (2005) 11967–11974.